Sanjeevani Twin · System Architecture

A hybrid physiology + AI digital twin for Type 2 Diabetes in India. It fuses a static EHR with dynamic wearable streams, predicts glucose spikes 2 h ahead, explains them and simulates interventions.

1 DATA SOURCES

Static · EHR (FHIR R4)
Demographics, diagnoses (SNOMED-CT), labs (LOINC: HbA1c, FPG, lipids, eGFR), meds (RxNorm), TCF7L2 genotype, polygenic risk, family history. Masked ABHA ID

Dynamic · wearables
CGM every 5 min · heart rate · HRV (RMSSD) · sleep stages · steps (Apple Health / Google Fit-like)

Patient logs
Meals from a 28-item Indian food library (GI-aware), medication doses

Synthetic cohort generator
40 Indian "virtual humans" × 14 days. Regional diets, Navratri fasting, late dinners, stress, sensor lag/noise/dropouts. Synthea-compatible

2 INGEST & HARMONISE

FHIR parser
LOINC/SNOMED/RxNorm → clinical profile; tolerant of partial records

Sleep episode detection
stage runs → hours, deep/REM fractions → sleep debt

Stress inference
HRV suppression + HR elevation beyond activity → latent stress 0–1

Adherence & habits
missed metformin doses → hepatic effect; usual meal slots learned in 3 days

Festival awareness
fasting foods in the log → expects sabudana/kuttu meals

3 DIGITAL TWIN CORE

EHR → Bayesian priors
BMI, waist (South-Asian weighting), age → S_I; duration + TCF7L2 → β-cell γ; FPG → G_b; sulfonylurea → threshold

Physiology engine
Extended Bergman minimal model: dual-speed gut absorption, exercise memory, circadian and sleep-debt modulation, stress-driven EGP, metformin effect

$dG/dt = -(S_G+X)G + S_G G_b' + Ra - k_{ex}AG + \ldots$

Calibration (onboarding, 3 days)
MAP cross-entropy search, 384 candidates × 3 rounds, batched NumPy, ~1 s/patient

Live assimilation + adaptive twin
Lag-compensated CGM observer every 5 min; nightly Bayesian re-calibration of S_I and hepatic set-point; learned dawn phenomenon

4 HYBRID AI LAYER

Habit-aware twin forecasts
simulate 120 min ahead, with and without the expected meal

Causal features (52)
twin state + forecasts + its own recent errors, CGM trend, meals, steps, HRV/stress, sleep, circadian, EHR. Leak-tested

Residual quantile GBMs
learn (CGM − twin) at 30/60/120 min · P10/P50/P90

Conformal calibration
CQR on disjoint calibration patients → honest 80% intervals

Spike classifier + isotonic calibration
P(new excursion above 180 within 2 h); alert threshold = max sensitivity at ≥75% precision

5 COUNTERFACTUAL REASONING

Shapley driver attribution
2^5 = 32 twin simulations: meal load, inactivity, sleep debt, stress, missed dose

What-if simulator
24 h of physiology under walks, low-GI swaps, earlier dinner, sleep, metformin dose

Prescription search
54 candidate plans in one batched run, ranked by benefit per unit of effort

Insights & nudges
personal glycaemic index for 28 foods, mined clinical brief, twin-tested advice in English and Hindi

6 CLINICIAN EXPERIENCE

Clinic triage
40 live twins ranked by predicted risk and time to event

3D holographic twin
organs glow with live state (insulin, gut, liver, HR, stress, muscle); glucose-coloured blood flow; data streams fuse in

Spike countdown + drivers
ETA, peak, interval, Shapley bars

Ghost twin · What-If Lab
two futures side by side in 3D, time-lapsed and synced to the chart; 54 futures visualised

FHIR in & out · Model card · Guided tour
RiskAssessment export, track record, validation transparency

Stdlib Python API
/cohort /patient /whatif /optimise /insights /fhir /fhir-out /metrics

Validation (10 held-out patients, 108 patient-days, never seen in training)

27.6	0.905	105 min	99.0%
mg/dL RMSE at 2 h (persistence 52.8)	spike AUROC (physics-only 0.80)	median warning time	Clarke zone A+B at 2 h

Why hybrid beats black-box ML

- 2 training patients: RMSE 31.3 vs 38.1 (ML-only), 18% better
- 26% fewer false alarms (0.65 vs 0.88/day) at higher precision (82% vs 79%)
- Explains and simulates. A plain predictor cannot

Privacy & standards

- Synthetic data only. DPDP Act 2023 / HIPAA-safe sandbox
- FHIR R4, LOINC, SNOMED-CT, RxNorm
- ABHA / ABDM-ready identifiers

Deployability

- CPU-only, runs on a laptop; no GPU or cloud
- Full pipeline in ~5 min; inference < 50 ms per patient
- Dependencies: numpy, scipy, pandas, scikit-learn

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